SharedPtr
SharedPtr
shared_ptr< smacc2::introspection::SmaccStateMachineInfo >
recursive_mutex
atomic< StateMachineInternalAction >
map< void *, std::shared_ptr< smacc2::CallbackCounterSemaphore > >
map< std::string, std::pair< std::function< std::string()>, boost::any > >
vector< smacc2_msgs::msg::SmaccTransitionLogEntry >
SMRunMode
int64_t
vector< smacc2::ISmaccState * >
shared_ptr< smacc2::introspection::SmaccStateInfo >
SmaccStatus
map< std::string, std::shared_ptr< smacc2::ISmaccOrthogonal > >
SharedPtr
#nh_
#timer_
#stateMachineInfo_
-eventQueueMutex_
-m_mutex_
-stateMachineCurrentAction
-stateCallbackConnections
-globalData_
-transitionLogHistory_
-runMode_
-stateSeqCounter_
#currentState_
#currentStateInfo_
#status_msg_
#orthogonals_
#eventsLogPub_
#stateMachinePub_
#stateMachineStatusPub_
#transitionHistoryService_
#transitionLogPub_
-statusPub_
smacc2::ISmaccStateMachine
+ ISmaccStateMachine()
+ ~ISmaccStateMachine()
+ reset()
+ stop()
+ eStop()
+ joinAsyncClientBehaviorThreads()
+ getOrthogonal()
+ getClientBehavior()
+ getOrthogonals()
+ requiresComponent()
+ requiresComponent()
+ postEvent()
+ postEvent()
+ getGlobalSMData()
+ setGlobalSMData()
+ getStateMachineName()
+ state_machine_visualization()
+ getCurrentStateInfo()
+ publishTransition()
+ onInitialize()
+ getTransitionLogHistory()
+ createSignalConnection()
+ disconnectSmaccSignalObject()
+ disconnectSmaccSignalObject()
+ notifyOnStateEntryStart()
+ notifyOnStateEntryEnd()
+ notifyOnRuntimeConfigured()
+ notifyOnStateExiting()
+ notifyOnStateExited()
+ notifyOnRuntimeConfigurationFinished()
+ getCurrentStateCounter()
+ getCurrentState()
+ getStateMachineInfo()
+ buildStateMachineInfo()
+ getNode()
+ getLogger()
+ getMutex()
# checkStateMachineConsistence()
# initializeROS()
# onInitialized()
# createOrthogonal()
- lockStateMachine()
- unlockStateMachine()
- propagateEventToStateReactors()
- updateStatusMessage()
int
weak_ptr< smacc2::introspection::SmaccStateMachineInfo >
weak_ptr< smacc2::introspection::SmaccStateInfo >
vector< smacc2::introspection::SmaccTransitionInfo >
vector< std::shared_ptr< smacc2::introspection::SmaccStateInfo > >
type_info *
std::enable_shared_from_this< SmaccStateInfo >
map< const std::type_info *, std::vector< smacc2::introspection::ClientBehaviorInfoEntry > >
map< const std::type_info *, std::vector< std::shared_ptr< smacc2::introspection::SmaccStateReactorInfo > > >
map< const std::type_info *, std::vector< std::shared_ptr< smacc2::introspection::SmaccEventGeneratorInfo > > >
+depth_
+stateIndex_
+stateMachine_
+parentState_
+transitions_
+children_
+tid_
+staticBehaviorInfo
+stateReactorsInfo
+eventGeneratorsInfo
+demangledStateName
+fullStateName
#node_
string
smacc2::introspection::SmaccStateInfo
+ SmaccStateInfo()
+ getStateLevel()
+ getStateMachineInfo()
+ getParentState()
+ depth()
+ getAncestors()
+ getFullPath()
+ createChildState()
+ declareTransition()
+ toShortName()
+ getDemangledFullName()
+ getNode()
+ getLogger()
vector< std::shared_ptr< smacc2::ISmaccClient > >
vector< std::vector< std::shared_ptr< smacc2::ISmaccClientBehavior > > >
#clients_
-clientBehaviors_
-stateMachine_
smacc2::ISmaccOrthogonal
+ setStateMachine()
+ getStateMachine()
+ initState()
+ addClientBehavior()
+ runtimeConfigure()
+ onEntry()
+ onExit()
+ onDispose()
+ getName()
+ requiresComponent()
+ requiresClient()
+ getClients()
+ getClientBehaviors()
+ setGlobalSMData()
+ getGlobalSMData()
+ getClientBehavior()
+ getNode()
+ getLogger()
# onInitialize()
# initializeClients()
# assignClientToOrthogonal()
- currentClientBehaviors()
mutex
atomic< bool >
vector< smacc2::ISmaccUpdatable * >
vector< std::vector< smacc2::ISmaccUpdatable * > >
atomic< int64_t >
double
SmaccFifoScheduler *
processor_handle
thread
ExecutionModel
-mutex_
-smaccStateMachine_
-signalDetector_
-end_
-initialized_
-rosInitialized_
-updatableClients_
-updatableStateElements_
-lastState_
-loop_rate_hz
-scheduler_
-processorHandle_
-signalDetectorThread_
-executionModel_
smacc2::SignalDetector
+ SignalDetector()
+ initialize()
+ setProcessorHandle()
+ runThread()
+ join()
+ stop()
+ terminateScheduler()
+ joinAsyncClientBehaviorThreads()
+ pollingLoop()
+ postEvent()
+ getNode()
+ getLogger()
+ notifyStateConfigured()
+ notifyStateExited()
+ notifyRosInitialized()
- pollOnce()
- findUpdatableClientsAndComponents()
- findUpdatableStateElements()
boost::signals2::signal< Signature, optional_last_value< typename boost::function_traits< Signature >::result_type >, int, std::less< int >, function< Signature >, typename extended_signature< function_traits< Signature >::arity, Signature >::function_type, boost::signals2::mutex >
shared_ptr< rclcpp::Logger >
vector< std::shared_ptr< smacc2::StateReactor > >
vector< std::shared_ptr< smacc2::SmaccEventGenerator > >
#stateInfo_
#logger_
#stateReactors_
#eventGenerators_
smacc2::ISmaccState
+ getStateMachine()
+ getName()
+ getParentState()
+ getNode()
+ getLogger()
+ getClassName()
+ configure()
+ requiresComponent()
+ requiresComponent()
+ requiresClient()
+ getGlobalSMData()
+ setGlobalSMData()
+ createStateReactor()
+ createStateReactor()
+ createEventGenerator()
+ postEvent()
+ postEvent()
+ notifyTransition()
+ notifyTransitionFromTransitionTypeInfo()
+ getStateReactors()
+ getEventGenerators()
+ getOrthogonal()
+ getClientBehavior()
+ getEventGenerator()
+ getStateReactor()
#parentState_
-stateMachine_
-orthogonal_
#stateMachine_
map< smacc2::ComponentKey, std::shared_ptr< smacc2::ISmaccComponent > >
#components_
smacc2::ISmaccClient
+ ISmaccClient()
+ ~ISmaccClient()
+ onInitialize()
+ getName()
+ getComponent()
+ getComponent()
+ getComponent()
+ getType()
+ getStateMachine()
+ connectSignal()
+ requiresClient()
+ getComponents()
+ iterateComponents()
+ postEvent()
+ postEvent()
# onComponentInitialization()
# onStateOrthogonalAllocation()
# createComponent()
# createNamedComponent()
# getNode()
# getLogger()
- initialize()
- setStateMachine()
- setOrthogonal()
-isShutdownRequested_
-mutex_
boost::signals2::signal< void(), optional_last_value< typename boost::function_traits< void()>::result_type >, int, std::less< int >, function< void ()>, typename extended_signature< function_traits< void()>::arity, void()>::function_type, boost::signals2::mutex >
boost::signals2::signal< void(const AprilTagDetectionArray::SharedPtr &), optional_last_value< typename boost::function_traits< void(const AprilTagDetectionArray::SharedPtr &)>::result_type >, int, std::less< int >, function< void(const AprilTagDetectionArray::SharedPtr &)>, typename extended_signature< function_traits< void(const AprilTagDetectionArray::SharedPtr &)>::arity, void (const AprilTagDetectionArray::SharedPtr &)>::function_type, boost::signals2::mutex >
smacc2::SmaccSignal< Signature, Combiner, Group, GroupCompare, SlotFunction, ExtendedSlotFunction, Mutex >
boost::signals2::signal< void(const AprilTagDetectionArray &), optional_last_value< typename boost::function_traits< void(const AprilTagDetectionArray &)>::result_type >, int, std::less< int >, function< void(const AprilTagDetectionArray &)>, typename extended_signature< function_traits< void(const AprilTagDetectionArray &)>::arity, void(const AprilTagDetectionArray &)>::function_type, boost::signals2::mutex >
boost::signals2::signal< void(const MessageType &), optional_last_value< typename boost::function_traits< void(const MessageType &)>::result_type >, int, std::less< int >, function< void(const MessageType &)>, typename extended_signature< function_traits< void(const MessageType &)>::arity, void(const MessageType &)>::function_type, boost::signals2::mutex >
-sub_
-currentOrthogonal
-currentState
-topicName_
-targetFrame_
-topicName_
< void()>
< void(const AprilTagDetectionArray::SharedPtr &)>
< void(const AprilTagDetectionArray &)>
< void(const MessageType &)>
smacc2::ISmaccClientBehavior
+ ISmaccClientBehavior()
+ ~ISmaccClientBehavior()
+ getStateMachine()
+ getName()
+ requiresClient()
+ requiresComponent()
+ requiresComponent()
+ onEntry()
+ onExit()
+ executeOnEntry()
+ executeOnExit()
# runtimeConfigure()
# postEvent()
# postEvent()
# getCurrentState()
# dispose()
# getNode()
# getLogger()
- onStateOrthogonalAllocation()
function< void()>
optional< std::future< int > >
smacc2::SmaccSignal< void()>
smacc2::SmaccSignal< void(const AprilTagDetectionArray::SharedPtr &)>
smacc2::SmaccSignal< void(const AprilTagDetectionArray &)>
optional< int >
smacc2::SmaccSignal< void(const MessageType &)>
function< void(const MessageType &)>
SharedPtr
bool
#owner_
smacc2::ISmaccComponent
+ ISmaccComponent()
+ ~ISmaccComponent()
+ getName()
# onInitialize()
# onComponentInitialization()
# postEvent()
# postEvent()
# onStateOrthogonalAllocation()
# requiresComponent()
# requiresComponent()
# requiresClient()
# createSiblingComponent()
# createSiblingNamedComponent()
# getNode()
# getLogger()
# getStateMachine()
- initialize()
- setStateMachine()
optional< std::string >
vector< std::string >
-postFailureEventFn_
-postFinishEventFn_
-postSuccessEventFn_
-onEntryThread_
-onExitThread_
-onFailure_
-onFinished_
-onSuccess_
+queueSize
+onFirstMessageReceived_
+onMessageReceived_
+postInitialMessageEvent
+postMessageEvent
-sub_
-firstMessage_
-initialized_
-selectedVisitTagId_
-visitedWorkingAreas_
smacc2::SmaccAsyncClientBehavior
+ onStateOrthogonalAllocation()
+ ~SmaccAsyncClientBehavior()
+ onSuccess()
+ onFinished()
+ onFailure()
+ requestForceFinish()
+ executeOnEntry()
+ executeOnExit()
+ waitOnEntryThread()
+ onSuccess()
+ onFinished()
+ onFailure()
# postSuccessEvent()
# postFailureEvent()
# dispose()
# isShutdownRequested()
- waitFutureIfNotFinished()
smacc2::client_core_components::CpTopicSubscriber< MessageType >
+ CpTopicSubscriber()
+ CpTopicSubscriber()
+ ~CpTopicSubscriber()
+ onMessageReceived()
+ onFirstMessageReceived()
+ onStateOrthogonalAllocation()
+ onInitialize()
- messageCallback()
function< void(const AprilTagDetectionArray &)>
cl_isaac_apriltag::CpAprilTagMissionState
+ CpAprilTagMissionState()
+ ~CpAprilTagMissionState()
+ onInitialize()
+ isTagVisited()
+ markTagVisited()
+ clearVisitedTags()
+ getVisitedTags()
+ selectTag()
+ getSelectedTag()
+ clearSelectedTag()
+onFirstMessageReceived
+onMessageReceived_
+queueSize
< AprilTagDetectionArray >
+postInitialMessageEvent
+postMessageEvent
-firstMessage_
-initialized_
smacc2::client_core_components::CpTopicSubscriber< AprilTagDetectionArray >
+ CpTopicSubscriber()
+ CpTopicSubscriber()
+ ~CpTopicSubscriber()
+ onMessageReceived()
+ onFirstMessageReceived()
+ onStateOrthogonalAllocation()
+ onInitialize()
- messageCallback()
shared_ptr< tf2_ros::Buffer >
shared_ptr< tf2_ros::TransformListener >
map< std::string, geometry_msgs::msg::PoseStamped >
+onAprilTagDetection_
-topicSubscriber_
-tfBuffer_
-tfListener_
-detectedAprilTagsMapPose
-missionState_
cl_isaac_apriltag::CpAprilTagTracker
+ CpAprilTagTracker()
+ ~CpAprilTagTracker()
+ onInitialize()
+ onComponentInitialization()
+ getTagsWithinTime()
+ getTagPose()
+ onAprilTagDetected()
- onAprilTagMessage()
-postEvAprilTagDetected_
-tracker_
cl_isaac_apriltag::CbDetectAprilTag
+ CbDetectAprilTag()
+ ~CbDetectAprilTag()
+ onOrthogonalAllocation()
+ onEntry()
+ onAprilTagDetected()